AWESOME CYBER SECURITY HEAVEN · REALM DOSSIER



The Proving Grounds

CTF Practice, Labs & Hacking Training

74	6	100%
FREE TOOLS	FLAGSHIP PICKS	FREE / OPEN

2026 Edition — every link re-verified 2026-10-15.
License-aware curation: copyleft, freemium and free-tier entries are tagged inline.
Part of **Awesome Cyber Security Heaven** — 438 curated tools across 14 realms.

ABOUT THIS REALM

The Proving Grounds — CTF Practice, Labs & Hacking Training

Hack legally — wargames, CTF platforms and deliberately-vulnerable apps to sharpen your skills.

Legal targets to practice on — vulnerable-by-design apps, CTF platforms, lab ranges. Practice targets are verified live before listing (dead practice apps are worse than none — they teach tool errors, not vulns).

- **Active in the last 30 days:** 8 of 32 tracked GitHub repos (25%)
- **Archived upstream (kept — still reference-quality):** [DVWS (Web Services)](https://github.com/snoopysecurity/dvws), [Hackademic](https://github.com/Hackademic/hackademic), [XVWA](https://github.com/s4n7h0/xvwa)
- **Quiet classics (no push in 2+ years, still the standard):** WackoPicko, OWASP NodeGoat, lfi-labs, 0l4bs, DVHMA, DVIA-v2, DVWS (Web Services), damnvulnerable.me, Hackademic, SentinelTestbed…
- **Youngest entries:** WackoPicko (2010), CryptOMG (2012), Hackademic (2012)

THE COLLECTION

All 74 tools — with live facts

Every entry below was checked against the GitHub API on 2026-10-15: star counts, project ages, last-push dates and licenses are re-snapshotted on every monthly verification pass. Flagship picks carry the **FLAGSHIP** badge — the tools this realm is built around. Everything is free, free-tier or open source: no paywalls, no SaaS wrappers.

Vulhub ★ FLAGSHIP

★ 21.4k · since 2017 · pushed 2026-10-14 · MIT

dockerized vulnerable environments: spin up a CVE, test, tear down.

`github.com/vulhub/vulhub`

OWASP Juice Shop ★ FLAGSHIP

★ 14.1k · since 2014 · pushed 2026-10-12

the modern deliberately-vulnerable app with gamified challenges covering the whole OWASP spectrum.

`github.com/juice-shop/juice-shop`

DVWA ★ FLAGSHIP

★ 13.8k · since 2013 · pushed 2026-10-07 · GPL-3.0

the classic Damn Vulnerable Web Application for safely learning injection basics.

`github.com/digininja/DVWA`

pwntools ★ FLAGSHIP

★ 13.8k · since 2013 · pushed 2026-09-03

the CTF exploit-development library: from sockets to ROP chains, pwn everything.

`github.com/Gallopsled/pwntools`

WebGoat ★ FLAGSHIP

★ 9.4k · since 2015 · pushed 2026-10-13 · GPL-2.0

OWASP's teaching app: guided lessons with a solution walkthrough for each vulnerability class.

`github.com/WebGoat/WebGoat`

GOAD ★ FLAGSHIP

★ 8.5k · since 2021 · pushed 2026-03-12 · GPL-3.0

Game of Active Directory: a three-forest lab purpose-built for safely practicing AD attack techniques.

`github.com/Orange-Cyberdefense/GOAD`

CTFd

★ 6.9k · since 2015 · pushed 2026-10-09

the CTF platform that powers thousands of events; self-host your own competition.

`github.com/CTFd/CTFd`

Metasploitable 3

★ 5.7k · since 2016 · pushed 2025-02-13 ⏳

Rapid7's purpose-built vulnerable target VM; the canonical Metasploit practice range.

`github.com/rapid7/metasploitable3`

CloudGoat

★ 3.8k · since 2018 · pushed 2026-10-14

Rhino Security's "vulnerable by design" AWS scenarios — deploy, attack, tear down.

`github.com/RhinoSecurityLabs/cloudgoat`

HEVD

★ 3.1k · since 2015 · pushed 2025-02-24 ⏳ · GPL-3.0

the Windows kernel-driver training target for exploit development.

`github.com/hacksysteam/HackSysExtremeVulnerableDrive`

APTSimulator

★ 2.8k · since 2018 · pushed 2025-09-23 ⏳ · MIT

makes a Windows host look compromised so you can test detections safely.

`github.com/NextronSystems/APTSimulator`

Splunk Attack Range

★ 2.6k · since 2019 · pushed 2026-10-13 · Apache-2.0

builds instrumented detection labs: attack, collect and test in one workflow.

`github.com/splunk/attack_range`

OWASP NodeGoat

★ 2.1k · since 2013 · pushed 2024-06-15 ⏳ · Apache-2.0

the Node.js vulnerable-by-design app: the Top 10 in a controllable lab.

`github.com/OWASP/NodeGoat`

XVWA

★ 1.8k · since 2015 · pushed 2020-09-12 ⏳ · GPL-3.0

Xtreme Vulnerable Web Application: dozens of flaws across one PHP app.

`github.com/s4n7h0/xvwa`

OWASP Mutillidae II

★ 1.5k · since 2018 · pushed 2026-09-24 · GPL-3.0

deliberately vulnerable app that re-creates every OWASP Top 10 issue, resettable per install.

`github.com/webpwnized/mutillidae`

flightsim

★ 1.4k · since 2018 · pushed 2024-04-04 ⏳

generates malicious network traffic for IDS/NDR validation; defender-side only. (CC-BY-4.0)

`github.com/alphasoc/flightsim`

DIVA Android

★ 1.2k · since 2016 · pushed 2023-05-19 ⏳ · GPL-3.0

Damn Insecure and Vulnerable App: the Android training classic.

`github.com/payatu/diva-android`

DVIA-v2

★ 1.1k · since 2018 · pushed 2024-03-29 ⏳ · MIT

Damn Vulnerable iOS App v2: the mobile testing curriculum as an app.

`github.com/prateek147/DVIA-v2`

AzureGoat

★ 966 · since 2022 · pushed 2024-10-30 ⏳ · MIT

the vulnerable-by-design Azure cloud lab.

`github.com/ine-labs/AzureGoat`

DSVW

★ 882 · since 2015 · pushed 2026-08-19

Damn Small Vulnerable Web: the Top 10 in ~100 lines of Python. (public domain)

`github.com/stamparm/DSVW`

Vulnerable Node

★ 492 · since 2016 · pushed 2026-10-06 · MIT-style

the Node.js vulnerable application for testing analysis tools.

`github.com/cr0hn/vulnerable-node`

DVWS (Web Services)

★ 461 · since 2016 · pushed 2021-12-06 ⏳ · Apache-2.0

vulnerable web services for SOA/API security practice.

`github.com/snoopysecurity/dvws`

DVWS (Web Sockets)

★ 363 · since 2017 · pushed 2026-07-24 · MIT

the vulnerable WebSocket app from Interference Security.

`github.com/interference-security/DVWS`

WackoPicko

★ 354 · since 2010 · pushed 2024-05-25 ⏳

the vulnerable app built for testing vulnerability scanners — a frozen-by-design calibration target for your tooling.

`github.com/adamdoupe/WackoPicko`

0l4bs

★ 350 · since 2020 · pushed 2021-06-02 ⏳ · GPL-3.0

XSS practice labs with progressive difficulty.

`github.com/tegal1337/0l4bs`

lfi-labs

★ 335 · since 2015 · pushed 2024-04-08 ⏳ · MIT

LFI/RFI/commix practice pages: one flaw family, many variants.

`github.com/paralax/lfi-labs`

Hackademic

★ 325 · since 2012 · pushed 2020-10-30 ⏳ · GPL-3.0

WordPress-based vulnerable lab scenarios for classroom use.

`github.com/Hackademic/hackademic`

Java Vulnerable Lab

★ 278 · since 2015 · pushed 2026-10-09 · GPL-2.0

the Java vulnerable web application from the CSPF crew.

`github.com/CSPF-Founder/JavaVulnerableLab`

DVHMA

★ 274 · since 2015 · pushed 2018-08-22 ⏳ · Apache-2.0

the Damn Vulnerable Hybrid Mobile App: Android+iOS appsec in one codebase.

`github.com/logicalhacking/DVHMA`

CryptOMG

★ 195 · since 2012 · pushed 2015-06-25 ⏳ · GPL-3.0

cryptographic flaw testbed from SpiderLabs.

`github.com/SpiderLabs/CryptOMG`

damnvulnerable.me

★ 34 · since 2015 · pushed 2019-05-19 ⏳ · MIT

a modern vulnerable webapp for the OWASP Top 10 era.

`github.com/skepticfx/damnvulnerable.me`

SentinelTestbed

★ 11 · since 2013 · pushed 2016-11-18 ⏳ · GPL-3.0

a testbed app for evaluating security instrumentation.

`github.com/dobin/SentinelTestbed`

Hack The Box

free tier

the platform for ranked machines, pro labs and active-dir ranges; the industry's proving ground.

`hackthebox.com`

TryHackMe

free tier

guided rooms from your first command to advanced red/blue paths — the friendliest on-ramp.

`tryhackme.com`

picoCTF

CMU's beginner-friendly CTF — the best first flag anyone can capture.

`picoctf.org`

OverTheWire

Bandit and friends: learn Linux, networking and protocols through SSH wargames.

`overthewire.org`

VulnHub

downloadable vulnerable VMs; build and break your own home lab offline.

`vulnhub.com`

CTFtime

the global CTF calendar, team rankings and writeup index — the hub of the sport.

`ctftime.org`

Cryptopals

the legendary crypto challenges; finish them and AES is no longer magic.

`cryptopals.com`

Gandalf

Lakera's free prompt-injection game; learn AI-security attacks hands-on, level by level.

`gandalf.lakera.ai`

bWAPP

the "buggy web application" with 100+ vulnerability flavors — an old-school playground that still teaches.

`itsecgames.com/`

Root Me

free tier

hundreds of real-world hacking challenges across dozens of categories, from web to stego.

`root-me.org/`

Google Gruyere

Google's cheese-themed vulnerable web app: exploit XSS, XSRF and information disclosure in the cloud.

google-gruyere.appspot.com/

CryptoHack

learn modern cryptography by breaking it: challenges from classical ciphers to ECC.

cryptohack.org/

Hackxor

a realistic web-app hacking game with a story arc — WebGoat with sharper teeth.

hackxor.net/

CyberDefenders

free tier

blue-team CTF platform: investigate realistic incidents with the tools a SOC actually uses.

cyberdefenders.org/

LetsDefend

free tier

browser-based SOC simulator: alert queues, playbooks and IR challenges mapped to MITRE ATT&CK.

letsdefend.io/

pwnable.kr

the classic system-hacking wargame: exploit your way from userland to kernel.

pwnable.kr/

Hack This Site

the community wargames veteran: missions, articles and forums since the early 2000s.

hackthissite.org/

Pentest-Ground

free service

hosted practice playground of intentionally vulnerable apps for security testing.

`pentest-ground.com/`

testasp.vulnweb.com

free service

the ASP demo target for scanner testing.

`testasp.vulnweb.com/`

testaspnet.vulnweb.com

free service

the ASP.NET demo target for scanner testing.

`testaspnet.vulnweb.com/`

Zero Bank

free service

demo banking site for web security exercises.

`zero.webappsecurity.com/`

Altoro Mutual

free service

the classic demo bank: legal target for appsec training and tool demos.

`demo.testfire.net/`

Firing Range

Apache-2.0

Google's XSS testbed: the original scanner benchmarking target.

`public-firing-range.appspot.com/`

XSS Game

free service

Google's XSS tutorial game: six levels from context basics to DOM tricks.

`xss-game.appspot.com/`

Arizona Cyber Warfare Range

free service

free live-fire ranges for defenders and researchers.

azcwr.org/az-cyber-warfare-ranges

SocketToMe

free service

Robin Wood's WebSocket testing challenge app.

digi.ninja/projects/sockettome.php

techlogia Lab

free browser-based hacking lab for security training. (CC-BY-4.0)

techlogia.de/en/lab

Crackmes.one

free service

the reverse-engineering challenge community: thousands of crackmes, all legal.

crackmes.one/

Microcorruption

free service

embedded-security CTF: reverse the lock firmware, level by level.

microcorruption.com/

MASTG Crackmes

the mobile RE challenges companion to the OWASP MASTG. (CC-BY-SA-4.0)

github.com/OWASP/mastg/tree/master/Crackmes

Reversing.kr

free service

the classic Korean reverse-engineering challenge site.

reversing.kr

Vulnserver

free service

the TCP vulnerable server for exploit-dev practice.

`thegreycorner.com/2010/12/introducing-vulnserver.htm`

CTFlearn

free service

the beginner-friendly CTF platform with a permanent challenge set.

`ctflearn.com/`

Hackertest.net

free service

the 20-level hacking simulation for web basics.

`hackertest.net/`

HackThis!!

free service

50+ levels of web security challenges, from basics to advanced.

`hackthis.co.uk/`

MysteryTwister C3

free service

the international cryptography challenge: from Caesar to quantum.

`mysterytwisterc3.org/en/`

RingZer0 CTF

free service

hundreds of challenges across web, crypto, forensics and reversing.

`ringzer0ctf.com/`

SmashTheStack

free service

the wargames network: shellcoding, exploitation and logic puzzles.

`smashthestack.org/`

W3Challs

free service

pentest training challenges with a ranking system.

`w3challs.com`

WeChall

free service

the challenge-site aggregator: one profile across all major wargames.

`wechall.net/`

ExploitMe Mobile

free service

the SecurityCompass mobile labs.

`securitycompass.github.io/AndroidLabs/index.html`

OWASP iGoat

wiki project

the iOS learning app from the OWASP mobile family.

`wiki.owasp.org/index.php/OWASP_iGoat_Tool_Project`



Keep this realm alive

Awesome Cyber Security Heaven is a hand-curated, license-verified map of 438 free cybersecurity tools across 14 realms — maintained, re-verified and free forever (CC0). Star the repo, propose a tool, or explore the other 13 realms.

Featured: 2MNY Website Vulnerability Scanner

200 free, self-hosted protection modules — SQLi, XSS, SSRF, subdomain takeover, CMS and AI-stack checks with evidence-backed, CVSS-scored reports. One Docker command, zero API keys.

github.com/RanaKhan77/awesome-cyber-security-heaven
2mny.com

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